

Solution Requirements

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (policymakers, researchers, students) must be able to securely register and log in using email/password or OAuth (Google/GitHub). Passwords must be securely hashed before storage.	High
2	Authorization levels	Define user roles: Admin (manages datasets, retrains models, manages users), Researcher/Policy maker (access to advanced analytics, batch uploads, and prediction history), and Student/Guest (basic single-country prediction access).	High

3	External interfaces	The system must integrate with the UNDP Human Development Reports API or World Bank Open Data API to automatically fetch and update baseline country statistics (Life Expectancy, Education, GNI).	Medium
4	Transactions processing	The system must process user-submitted country indicators, execute the Machine Learning inference pipeline, and log the transaction (prediction result) into the user's history database without data loss or duplication.	High
5	Reporting	The system must generate visual dashboards (using Matplotlib/Seaborn) showing the predicted HDI tier, comparative analysis with global averages, and allow users to export results as PDF or CSV files.	High
6	Business rules, etc.	The ML model must classify countries into four tiers based on standard UNDP thresholds: Very High (≥ 0.800) , High (≥ 0.700 to < 0.800) , Medium (≥ 0.550 to < 0.700) , and **Low (< 0.550)** . Input validation must enforce realistic ranges (e.g., Life Expectancy 0-100, GNI > 0).	High
7	Compliance to laws or regulations	The system must comply with general data protection regulations (e.g., GDPR) regarding the secure storage, processing, and right-to-deletion of user Personally Identifiable Information (PII) like emails and names.	Medium
8	Other	The system must support batch predictions , allowing users to upload a CSV file containing multiple countries' data for bulk processing and downloading the results.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The ML prediction inference and web page rendering must be highly responsive to ensure a smooth user experience without noticeable lag.	ML prediction inference time < 1 second . Web page load time < 2 seconds .
2	Scalability	The system architecture must handle growth in concurrent users and expanding historical datasets (adding more years and new countries) without performance degradation.	Supports up to 500 concurrent users . Handles datasets up to 50 years of data for 190+ countries .
3	Security & Data Privacy	All user data and API endpoints must be protected against common web vulnerabilities (SQL injection, XSS). Data in transit must be encrypted.	Passwords hashed using bcrypt . Data transmitted over TLS 1.2/1.3 . ORM used to prevent SQL injection.
4	Reliability & Availability	The application must be highly available and include fault tolerance. If the primary ML model fails, the system should gracefully degrade or show a user-friendly error.	99.5% uptime per month. Automated fallback mechanism with < 5% error rate on valid inputs.
5	Usability & Accessibility	The user interface must be intuitive for non-technical users (like policymakers) and accessible to users with disabilities.	Responsive UI (mobile/desktop). Complies with WCAG 2.1 AA standards for color contrast and screen readers.
6	Other (Maintainability & Portability)	The codebase must be modular, well-documented, and easily deployable across different environments (local, cloud) without complex configuration	Code follows PEP-8 standards . System is containerized using Docker for seamless deployment on AWS/Azure/Heroku.